COMPUTER SYSTEMS FUNDAMENTALS

Hardware Overview & Architecture

BCA - First Semester

Computer System Overview

Definition : A computer system is an integrated set of components for input, processing, and output involving the user, hardware, and software working together to convert data into useful information.

How it works:

- Accept data and instructions from users (Input)
- CPU processes data using operations
- Display/store processed results (Output)

IPO Cycle



Example: Calculating student grades

- Input: Student marks
- Process: Calculate average, assign grade
- Output: Grade report

Input-Process-Output (IPO) Model

Definition : The IPO model represents the fundamental operation of a computer: accepting raw data as input, processing it through computational operations, and producing processed results as output.

INPUT

Raw data from users:

- Keyboard input
- Mouse clicks
- Scanned documents
- Sensors/Microphone

PROCESS

CPU operations:

- CPU executes instructions
- ALU performs calculations
- Memory stores intermediate results
- Logic decisions made

OUTPUT

Processed results:

- Monitor display
- Printed document
- Audio output
- File storage

Data vs Information

DATA :

Raw facts and figures with no relational meaning. Collection of symbols, numbers, or characters.

INFORMATION :

Processed, organized, and interpreted data that provides meaning and is useful for decision-making.

Comparison:

DATA	INFORMATION
Raw unprocessed facts	Processed and interpreted facts
Example: 90, 85, 78, 92	Example: Average score = 86.25
No specific meaning	Has specific meaning & context
Hard to make decisions	Easy to make decisions

Performance Basics

Definition : Computer performance is measured by the ability to accomplish a defined amount of work in a given time interval, balancing speed, throughput, and resource efficiency.

Execution Time

Time to complete one task

Shorter is better. Measured in seconds.

Throughput

Number of tasks completed per second

More tasks done is better.

Other Important Performance Metrics:

Clock Speed (GHz) Frequency of CPU cycles. Higher frequency = faster execution, but more heat & power

MIPS Million Instructions Per Second. Indicates how many instructions CPU can execute per second

CPI Cycles Per Instruction. Average number of clock cycles needed to execute one instruction

Bandwidth Maximum rate of data transfer between CPU, memory, and I/O devices

- **CLOCK SPEED :** when u ride a bicycle do u reach the destination fast? / chef cooking very fast
  yes, same way CPU works in very tiny steps called **clock cycles.**

NOTE : Higher clock cycle =/ always faster [also depends on processor design]

- **MIPS:** **Workers :** A -> 100 boxes per hr
  B -> 200 per hr : Worker B is faster

Note : More instructions every sec = better performance

Chef prepares many dishes every min (High MIPS)

- **CPI :Lower Cpi = better**
  Imagine writing one sentence -> A -> 3 min
  -> B -> 1 min

Efficient ? = B
CPU needs – 5 clock cycle and 2 clock cycle : 2 cycle is better as it finishes the instruction fast

[Lower CPI = Better]

(Chef needs only 2 steps to finish the dish)

CPU Components

ALU - Arithmetic Logic Unit :

Definition: Part of the CPU that performs arithmetic operations (addition, subtraction, multiplication, division) and logical operations (AND, OR, NOT).

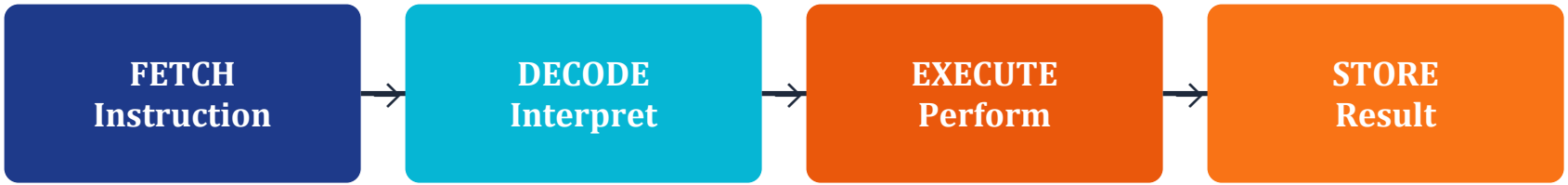
[It's like a calculator that does math and makes logic decisions.]

CU - Control Unit :

Definition: Part of the CPU responsible for fetching instructions from memory, decoding them, and directing the execution of operations.

[It's the manager that reads instructions and tells the ALU what to do.]

CPU Execution Process:



Registers :
Why are they needed?
Suppose ALU is calculating 100 + 200
Where should those numbers stay?
Inside Registers.
Registers temporarily hold : Numbers , Addresses, Instructions , Results

Analogy
Imagine cooking.
You keep : Salt , Oil , Masala on the kitchen table. [Not inside the cupboard.]
Because you need them immediately.
Registers work the same way.

"What is the brain of the computer?"

They will answer: **CPU**

Then ask: "Does the brain do only one job?"

No.

Our brain :

- Thinks
- Remembers
- Makes decisions
- Controls our body

Similarly, the CPU also has different parts.

What is CPU?

CPU stands for

Central Processing Unit

It is called the **Brain of the Computer** because every instruction passes through it.

Examples

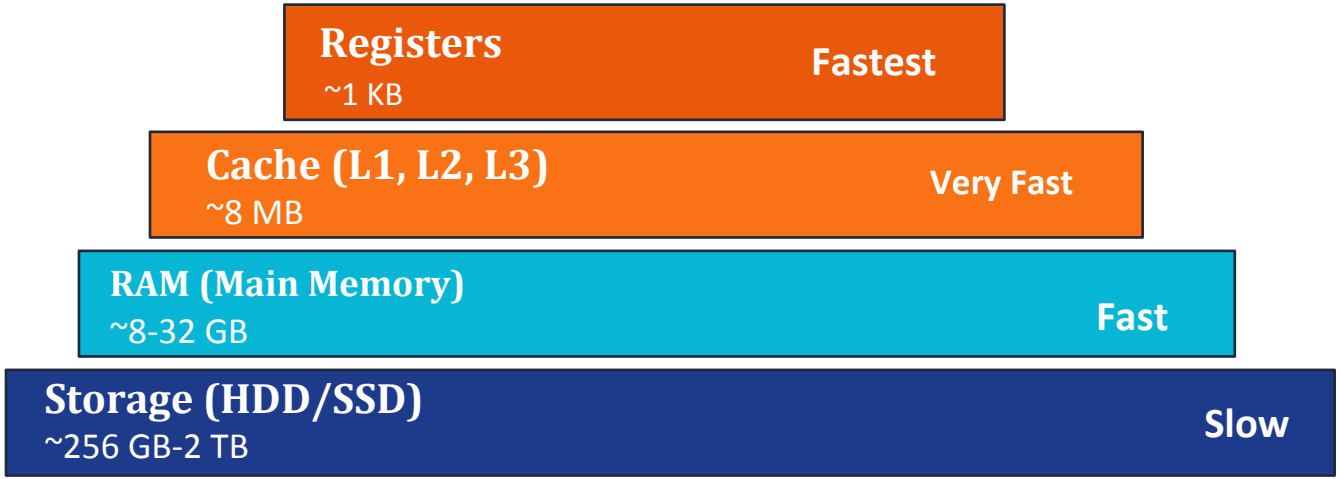
When you

- Open Chrome
- Play YouTube
- Save a file
- Type in Word
- Open WhatsApp

The CPU performs all these operations.

Memory Hierarchy

Definition : Memory hierarchy is an arrangement of memory devices in decreasing speed and increasing capacity, with a tradeoff between cost and performance.



Core Principle: CPU checks registers → cache → RAM for data (speed), and stores less-used data in storage (capacity)

Analogy: Like having a Pocket (registers), backpack (cache), desk (RAM), and filing cabinet (storage) for documents - you keep frequently needed items close!

Name	Register	Cache	Main Memory	Secondary Memory
Size	< 1 Kb	<16 Mb	<16 Gb	>100Gb
Access Time	0.25ns to 0.5ns	0.5ns to 25ns	80ns to 250ns	50,000ns
Managed by	Compiler	Hardware	Operating System	Operating System
Backing Mechanism	From Cache	From Main Memory	From Secondary Memory	Nothing (Sometimes I/O like pendrive/CD)

Memory	Speed	Size	Volatile?
Registers	Fastest	Very Small	Yes
Cache	Very Fast	Small	Yes
RAM	Fast	Medium	Yes
SSD	Slow compared to RAM	Large	No
HDD	Slowest	Very Large	No

Storage & I/O Devices

Storage Technologies Comparison:

HDD (Hard Disk Drive)

Definition: Magnetic storage with rotating platters and read/write heads.

Characteristics:

✓ Large capacity (1-10 TB)
✗ Moving mechanical parts
✗ Slower (~5-7K RPM)
✓ Low cost per GB

SSD (Solid State Drive)

Definition: Electronic storage using flash memory chips; no moving parts.

Characteristics:

✓ Very fast speed
✓ No moving parts
✓ Energy efficient
✗ Higher cost per GB

I/O Devices :

Definition: Hardware devices that enable communication between computer and user/external systems.

- **Input:** Keyboard, Mouse, Scanner
- **Output:** Monitor, Printer, Speaker
- **Bidirectional:** Touchscreen, USB

I/O Controller :

Definition: Hardware component that manages data transfer between CPU and I/O devices, handling protocol conversion and timing.

Simple: Translates between fast CPU and slower devices.

HDD	SSD
Rotating Disk	Flash Chips
Mechanical	Electronic
Slow	Fast
Cheap	Costly
Heavy	Light
Noisy	Silent

Component	Real-Life Analogy
CPU	Brain
ALU	Calculator
Control Unit	Principal / Manager
Registers	Money in Hand
Cache	Wallet
RAM	School Bag
SSD/HDD	Cupboard
Keyboard	Input
Monitor	Output
I/O Controller	Translator

Then ask students one by one:

- "Which part is like the calculator?" → **ALU**
- "Which memory is like a school bag?" → **RAM**
- "Which component is like the principal?" → **Control Unit**
- "Which storage is permanent?" → **SSD/HDD**
- "Which memory is the fastest?" → **Registers**
- "What happens if RAM loses power?" → **Its contents are lost because RAM is volatile.**

Summary

- Computer systems follow Input-Process-Output model
- Data becomes Information through processing
- CPU (ALU + CU) executes all instructions
- Memory Hierarchy balances speed vs. capacity
- Storage options: HDD (cheap/large) vs SSD (fast/reliable)
- I/O devices enable human-computer interaction

Understanding hardware is essential for programming and system design!